

||Jai Sri Gurudev||

S. J. C. INSTITUTE OF TECHNOLOGY, CHICKBALLAPUR

Department of Mathematics

LECTURE NOTES

Advanced Calculus and Numerical Methods (21MAT21)

Module-5 (Numerical Solution of Ordinary Differential Equations)

Prepared By:

Purushotham.P

Assistant Professor

SJC Institute of Technology

Email id: purushotham@sjcit.ac.in

Phone: 7338005244

Module - 05

Taylor's series method:-

step-1:- Write the given D.E into the standard form form

$\frac{dy}{dx} = y'(x) = f(x, y)$ to the initial condition, $y(x_0) = y_0$.

step-2:- Identify $y'(x)$, x_0 , y_0 and find $y'(x_0)$.

step-3:- Differentiate $y'(x)$ successively with respect to 'x' and obtain $y''(x)$, $y'''(x)$, $y^{IV}(x)$ -----

step-4:- Find $y''(x_0)$, $y'''(x_0)$, $y^{IV}(x_0)$ ---- and substitute all the above values in the Taylor's series formula given below.

$$y(x) = y(x_0) + \frac{(x-x_0)^1}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \dots$$

1. Solve $\frac{dy}{dx} = x + y$, to the initial condition $y(0) = 1$ using Taylor series method and thence find $y(0.1)$.

Sol:- Given $\frac{dy}{dx} = x + y = y'(x)$
 $\Rightarrow x_0 = 0, y_0 = 1$

$$\therefore y'(x_0) = x_0 + y_0 = 0 + 1 = 1$$

$$\begin{aligned}\therefore y''(x_0) &= 1 + y' \Rightarrow y''(x_0) = 1 + y'_0 \\ &= 1 + 1 \\ &\Rightarrow y''(x_0) = 2\end{aligned}$$

$$\therefore y'''(x) = 0 + y'' \Rightarrow y'''(x_0) = 2$$

$$\therefore y^{IV}(x) = y''' \Rightarrow y^{IV}(x_0) = y'''_0 = 2$$

$\therefore$ WKT,

$$y(x) = y_0 + \frac{(x-x_0)^1}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \dots$$

$$\Rightarrow y(x) = 1 + \frac{x}{1!} (1) + \frac{x^2}{2!} (2) + \frac{x^3}{3!} (2) + \frac{x^4}{4!} (2) + \dots$$

$$\Rightarrow y(x) = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \dots$$

$$\Rightarrow y(0.1) = 1 + (0.1) + (0.1)^2 + \frac{(0.1)^3}{3} + \frac{(0.1)^4}{12} + \dots$$

$$\Rightarrow y(0.1) = 1 + 0.1 + 0.001 + 0.0003 + 0$$

$$\Rightarrow y(0.1) = 1.1013$$

Q. Using Taylor's series method, find the appropriate value of $y(0.1)$ from $\frac{dy}{dx} = 3x + y^2$, with $y(0) = 1$.

Sol:- Given $\frac{dy}{dx} = 3x + y^2 = y'(x) \rightarrow (1)$

and $y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$

$$\therefore y(x_0) = 3x_0 + y_0^2 = 3(0) + 1^2 = 1 = y'_0$$

$$\therefore y'(x) = 3 + 2yy'$$

$$\Rightarrow y'(x_0) = 3 + 2y_0 y'_0$$

$$\Rightarrow y''(x_0) = 3 + 2(1)(1) = 5$$

$$\therefore y'''(x) = 0 + 2[y'y' + y y'']$$

$$\Rightarrow y'''(x) = 2[y'^2 + y y'']$$

$$\Rightarrow y'''(x_0) = 2[(y'_0)^2 + y_0 y''_0]$$

$$\Rightarrow y'''(x_0) = 2[(1)^2 + (1)(5)]$$
$$= 2(6)$$

$$\Rightarrow y'''(x_0) = 12$$

WKT,

$$y(x) = y_0 + \frac{(x-x_0)^1}{1!} y'_0 + \frac{(x-x_0)^2}{2!} y''_0 + \frac{(x-x_0)^3}{3!} y'''_0 + \dots$$

$$\Rightarrow y(x) = 1 + \frac{x}{1!}(1) + \frac{x^2}{2!}(5) + \frac{x^3}{3!}(12) + \dots$$

$$\Rightarrow y(x) = 1 + x + \frac{5}{2} x^2 + 2x^3 + \dots$$

$$\Rightarrow y(x) = 1 + 0.5 + \frac{5}{2} (0.1)^2 + 2(0.1)^3$$

$$\Rightarrow y(0.1) = 1 + 0.5 + 0.025 + 0.002$$

$$\Rightarrow y(0.1) = 1.1270$$

3. Using Taylor's series method $y(0.2)$ from $\frac{dy}{dx} = x^2y - 1$ with $y(0) = 1$.

Given, $\frac{dy}{dx} = x^2y - 1 = y'(x) \rightarrow (1)$

$$\text{and } y(0) = 1, \Rightarrow x^2y - 1 = y'(x) \\ \Rightarrow x_0 = 0, y_0 = 1$$

$$\therefore y'(x_0) = x_0^2 y_0 - 1 = 0(1) - 1 = -1$$

$$y''(x) = 2xy + x^2 y' - 0$$

$$\Rightarrow y''(x_0) = 2x_0 y_0 + x_0^2 y'_0 = 2(0)(1) + (0)(-1) = 0$$

$$\therefore y'''(x) = 2[y + xy'] + [2xy' + x^2 y'']$$

$$\Rightarrow y'''(x_0) = 2[y_0 + x_0 y'_0] + [2x_0 y'_0 + x_0^2 y''_0]$$

$$\Rightarrow y'''(x_0) = 2[1 + 0] + [0 + 0] = 2$$

$$\therefore y^{(4)}(x) = 2[y' + y' + xy''] + [2y' + 2xy'' + 2xy'' + x^2 y''']$$

$$\Rightarrow y^{(4)}(x) = 2[2y' + xy''] + [2y' + 4xy'' + x^2 y''']$$

$$\Rightarrow y^{(4)}(x_0) = 2[2y'_0 + x_0 y''_0] + [2y'_0 + 4x_0 y''_0 + x_0^2 y'''_0]$$

$$\Rightarrow y^{(4)}(x_0) = 2[-2 + 0] + [-2 + 0 + 0]$$

$$\Rightarrow y^{(4)}(x_0) = -4 - 2 = -6$$

$\therefore$ WKT,

$$y(x) = y_0 + \frac{(x-x_0)^1}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \dots$$

$$\Rightarrow y(x) = 1 + \frac{x}{1!} (-1) + \frac{x^2}{2!} (0) + \frac{x^3}{3!} (2) + \frac{x^4}{4!} (-6) + \dots$$

$$\Rightarrow y(x) = 1 - x + \frac{1}{3} x^3 - \frac{1}{4} x^4 + \dots$$

$$\Rightarrow y(0.2) = 1 - 0.2 + \frac{1}{3} (0.2)^3 - \frac{1}{4} (0.2)^4$$

$$\Rightarrow y(0.2) = 1 - 0.2 + 0.0027 - 0.0004 = 0.8023$$

4. Employ Taylor series method, evaluate $y(0.2)$ from $\frac{dy}{dx} = e^x - y^2$ with $y(0) = 1$.

Sol: Given, $\frac{dy}{dx} = e^x - y^2 = y'(x) \rightarrow (1)$

and $y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$.

$$\therefore y'(x_0) = e^{x_0} - y_0^2 = e^0 - 1^2 = 1 - 1 = 0$$

$$\therefore y''(x) = e^x - 2yy'$$

$$\Rightarrow y''(x_0) = e^{x_0} - 2y_0 y'_0 = e^0 - 2(1)(0) = 1$$

$$\therefore y'''(x) = e^x - 2[y'y' + yy'']$$

$$\begin{aligned}\Rightarrow y'''(x_0) &= e^{x_0} - 2[y'_0 y'_0 + y_0 y''_0] \\ &= e^0 - 2[0 + (1)(1)] = 1 - 2 = -1\end{aligned}$$

$$\therefore y^{IV}(x) = e^x - 2[2y'y'' + y'y''' + yy^{IV}]$$

$$\Rightarrow y^{IV}(x) = e^x - 2[3y'y'' + yy''']$$

$$\Rightarrow y^{IV}(x_0) = e^{x_0} - 2[3y'_0 y''_0 + y_0 y'''_0]$$

$$\Rightarrow y^{IV}(x_0) = e^0 - 2[3(0)(1) + (1)(-1)]$$

$$= 1 + 2$$

$$\Rightarrow y^{IV}(x_0) = 1 + 2 = 3$$

WKT,

$$y(x) = y_0 + \frac{(x-x_0)^1}{1!} y'_0 + \frac{(x-x_0)^2}{2!} y''_0 + \frac{(x-x_0)^3}{3!} y'''_0 + \dots$$

$$\Rightarrow y(x) = 1 + \frac{x}{1!}(0) + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(-1) + \frac{x^4}{4!}(3) + \dots$$

$$\Rightarrow y(x) = 1 + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{8} + \dots$$

$$\therefore y(0.2) = 1 + \frac{(0.2)^2}{2} - \frac{(0.2)^3}{6} + \frac{1}{8}(0.2)^4 = 1.0189$$

5. Solve $y' = x^2y + 1$ with $y(0) = 0$ and hence find the value of y at $x = 0.4$ using Taylor series method.

sol: Given $y' = x^2y + 1$

$$\Rightarrow \frac{dy}{dx} = x^2y + 1 = y'(x) \rightarrow (1)$$

$$\text{and } y(0) = 0 \Rightarrow x_0 = 0, y_0 = 0$$

$$\therefore y'(x_0) = 0(0) + 1 = 1$$

$$\therefore y''(x) = x^2y' + y(2x) + 0$$

$$\Rightarrow y''(x) = x^2y' + 2xy$$

$$\Rightarrow y''(x_0) = x_0^2y'_0 + 2x_0y_0 = (0)(0) + 2(0)(0) = 0$$

$$\therefore y'''(x) = x^2y'' + y'(2x) + 2[xy' + y]$$

$$\Rightarrow y'''(x) = 4xy' + 2y + x^2y''$$

$$\Rightarrow y'''(x_0) = 4x_0y'_0 + 2y_0 + x_0^2y''_0 = 4(0)(0) + 0 + 0 = 0$$

$$\therefore y^{(4)}(x) = 6xy'' + 6y' + x^2y'''$$

$$\begin{aligned} \Rightarrow y^{(4)}(x_0) &= 6x_0y''_0 + 6y'_0 + x_0^2y'''_0 \\ &= 6(0)(0) + 6(1) + (0)(0) = 0 + 6 + 0 \\ &= 6 \end{aligned}$$

WIKT,

$$y(x) = y(x_0) + \frac{(x-x_0)^1}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \dots$$

$$\Rightarrow y(x) = 0 + x(1) + \frac{x^2}{2} (0) + \frac{x^3}{6} (0) + \frac{x^4}{24} (6)$$

$$\Rightarrow y(x) = x + \frac{1}{4} x^4$$

$$\Rightarrow y(0.4) = 0.4 + (0.25)(0.4)^4$$

$$\Rightarrow y(0.4) = 0.4 + 0.0064$$

$$\Rightarrow y(0.4) = 0.4064$$

6. Solve $\frac{dy}{dx} = e^x - y$ with $y(0) = 1$ & hence find $y(0.1)$ using Taylor series method.

Sol: Given $\frac{dy}{dx} = e^x - y = y'(x) \rightarrow (1)$

& $y(0) = 1, \Rightarrow x_0 = 0, y_0 = 1$

$\therefore y'(x) = e^x - y$

$\Rightarrow y'(x_0) = e^{x_0} - y_0 = e^0 - 1 = 1 - 1 = 0$

$\therefore y''(x) = e^x - y'$

$\Rightarrow y''(x_0) = e^{x_0} - y'_0 = 1 - 0 = 1$

$\therefore y'''(x) = e^x - y''$

$\Rightarrow y'''(x_0) = e^{x_0} - y''_0 = e^0 - 1 = 1 - 1 = 0$

$\therefore y^{IV}(x) = e^x - y'''$

$\Rightarrow y^{IV}(x_0) = e^{x_0} - y'''_0 = e^0 - 0 = 1 - 0 = 1$

WKT,

$\Rightarrow y(x) = y(x_0) + \frac{(x-x_0)^1}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \dots$

$\Rightarrow y(x) = 1 + x(0) + \frac{x^2}{2} (1) + \frac{x^3}{6} (0) + \frac{x^4}{24} (1) + \dots$

$\Rightarrow y(x) = 1 + \frac{x^2}{2} + \frac{x^4}{24}$

$\Rightarrow y(0.1) = 1 + \frac{(0.1)^2}{2} + \frac{(0.1)^4}{24}$

$\Rightarrow y(0.1) = 1 + 0.005 + 0$

$\Rightarrow y(0.1) = 1.005$

4. Employ Taylor series method to find y at $x=0.1$ correct to 4 decimal places, given $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$.

Sol:- Given, $\frac{dy}{dx} = 2y + 3e^x = y'(x) \rightarrow (1)$

and $y(0) = 0 \Rightarrow x_0 = 0, y_0 = 0$

$$\Rightarrow y'(x_0) = 2y_0 + 3e^{x_0} = 2(0) + 3e^0 = 3$$

$$\therefore y''(x) = 2y' + 3e^x$$

$$\Rightarrow y''(x_0) = 2y'_0 + 3e^{x_0} = 2(3) + 3 = 9$$

$$\therefore y'''(x) = 2y'' + 3e^x$$

$$\Rightarrow y'''(x_0) = 2y''_0 + 3e^{x_0} = 2(9) + 3(1) = 21$$

$$\therefore y^{IV}(x) = 2y''' + 3e^x$$

$$\Rightarrow y^{IV}(x_0) = 2y'''_0 + 3e^{x_0} = 2(21) + 3 = 42 + 3 = 45$$

WKT,

$$y(x) = y(x_0) + \frac{(x-x_0)^1}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \dots$$

$$\Rightarrow y(x) = 0 + (x-0)(3) + \frac{(x-0)^2}{2} (9) + \frac{(x-0)^3}{6} (21) + \frac{(x-0)^4}{24} (45)$$

$$\Rightarrow y(x) = 3x + \frac{9x^2}{2} + \frac{21x^3}{6} + \frac{15x^4}{24}$$

$$\Rightarrow y(x) = 3x + \frac{9}{2} x^2 + \frac{7}{2} x^3 + \frac{15}{8} x^4$$

$$\Rightarrow y(0.1) = 3(0.1) + \frac{9}{2} (0.1)^2 + \frac{7}{2} (0.1)^3 + \frac{15}{8} (0.1)^4$$

$$\Rightarrow y(0.1) = 0.3 + 0.045 + 0.0035 + 0.0002$$

$$\Rightarrow y(0.1) = 0.3487.$$

Modified Euler's Method:-

Step-1: Write the given D.E into the standard form

$$\frac{dy}{dx} = f(x, y) \text{ to the initial condition } y(x_0) = y_0$$

Step-2:- To find $y(x_1) = y_1$:-

$$I - (1) : y(x_0 + h) = y(x_1) = y_1^{(1)} = y_0 + hf(x_0, y_0)$$

$$I - (2) : y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$I - (3) : y_1^{(3)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})]$$

$$I - (4) : y_1^{(4)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(3)})]$$

Continue the same process until to reach accuracy value of y . i.e., $y(x_1) = y_1$

Step-3:- To find $y(x_2) = y_2$:-

$$I - (1) : y(x_1 + h) = y(x_2) = y_2^{(1)} = y_1 + hf(x_1, y_1)$$

$$I - (2) : y_2^{(2)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})]$$

$$I - (3) : y_2^{(3)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(2)})]$$

Continue the same process until to reach the accuracy value of y i.e., $y(x_2) = y_2$.

1. Use modified Euler's method to compute $\frac{dy}{dx} = 3x + \frac{y}{2}$
 $y(0) = 1$, perform three iterations by taking $h = 0.1$.

Sol:- Given, $\frac{dy}{dx} = 3x + \frac{y}{2} = f(x, y)$

and $y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$ & $h = 0.1$

To find $y(x_1) = y_1$:-

$$\therefore y(x_0 + h) = y(x_1) = y_1^{(1)} = y_0 + h f(x_0, y_0)$$

$$\Rightarrow y_1^{(1)} = 1 + (0.1) f(0, 1)$$

$$\Rightarrow y_1^{(1)} = 1 + (0.1) (0.5)$$

$$\Rightarrow y_1^{(1)} = 1 + 0.05$$

$$\Rightarrow y_1^{(1)} = 1.05$$

$$\therefore y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.05)]$$

$$= 1 + 0.05 [0.5 + 0.825]$$

$$\Rightarrow y_1^{(2)} = 1.06625$$

$$\therefore y_1^{(3)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})]$$

$$= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.06625)]$$

$$= 1 + 0.05 [0.5 + 0.833125]$$

$$\Rightarrow y_1^{(3)} = 1.0666$$

$$\therefore y_1^{(4)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(3)})]$$

$$= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.0666)]$$

$$= 1 + 0.05 [0.5 + 0.8333]$$

$$\Rightarrow y_1^{(4)} = 1.0666$$

$$\therefore y(x_1) = y(0.1) = 1.0666$$

Q. Use modified Euler's Method, find $y(0.1)$ taking $h = 0.05$, given that $\frac{dy}{dx} = x^2 + y$ with $y(0) = 1$.

Sol. Given, $\frac{dy}{dx} = x^2 + y = f(x, y)$

and $y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1, h = 0.05$

To find $y(x_1) = y_1$:-

$$\begin{aligned}\therefore y(x_0 + h) &= y_1 = y_0 + h f(x_0, y_0) \\ &= 1 + 0.05 f(0, 1) \\ &= 1 + (0.05)(1)\end{aligned}$$

$$\Rightarrow y_1^{(1)} = 1.05$$

$$\begin{aligned}\therefore y_1^{(2)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})] \\ &= 1 + \frac{0.05}{2} [f(0, 1) + f(0.05, 1.05)]\end{aligned}$$

$$= 1 + 0.025 [1 + 1.0525]$$

$$\Rightarrow y_1^{(2)} = 1.0513$$

$$\begin{aligned}\therefore y_1^{(3)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})] \\ &= 1 + \frac{0.05}{2} [f(0, 1) + f(0.05, 1.0513)]\end{aligned}$$

$$= 1 + 0.025 [1 + 1.0538]$$

$$\Rightarrow y_1^{(3)} = 1.0513$$

$$\Rightarrow x_1 = 0.05, y_1 = 1.0513$$

To find $y(x_2) = y_2$:-

$$\begin{aligned}\therefore y(x_1 + h) &= y(x_2) = y_2^{(1)} = y_1 + h f(x_1, y_1) \\ &= 1.0513 + (0.05) f(0.05, 1.0513)\end{aligned}$$

$$= 1.0513 + (0.05)(1.0538)$$

$$\Rightarrow y_2^{(1)} = 1.1040.$$

$$\therefore y_2^{(2)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})]$$

$$= 1.0513 + \frac{0.05}{2} [f(0.05, 1.0513) + f(0.1, 1.104)]$$

$$= 1.0513 + 0.025 [1.0538 + 1.114]$$

$$\Rightarrow y_2^{(2)} = 1.1054.$$

$$\therefore y_2^{(3)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(2)})]$$

$$= 1.0513 + \frac{0.05}{2} [f(0.05, 1.0513) + f(0.1, 1.1054)]$$

$$= 1.0513 + 0.025 [1.0538 + 1.1154]$$

$$\Rightarrow y_2^{(3)} = 1.1055$$

$$\therefore y(x_2) = y(0.1) = 1.1055.$$

3. By using modified Euler's method, find $y(0.2)$, taking $h=0.1$, given from $\frac{dy}{dx} = \frac{y-x}{y+x}$ this $y(0)=1$.

Sol: Given $\frac{dy}{dx} = \frac{y-x}{y+x} = f(x, y)$

$$y(0)=1 \Rightarrow x_0=0, y_0=1, h=0.1$$

To find $y(x_1) = y_1$:-

$$\therefore y(x_0+h) = y_1^{(1)} = y_0 + hf(x_0, y_0)$$

$$= 1 + 0.1f(0, 1)$$

$$= 1 + (0.1)(1)$$

$$\Rightarrow y_1^{(1)} = 1.1$$

$$\therefore y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.1)]$$

$$= 1 + 0.05 [1 + 0.8333]$$

$$\Rightarrow y_1^{(2)} = 1.0916$$

$$\therefore y_1^{(3)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})]$$

$$= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.0916)]$$

$$= 1 + 0.05 [1 + 0.83315]$$

$$\Rightarrow y_1^{(3)} = 1.0916$$

$$\therefore y(0.1) = 1.0916, \quad x_1 = 0.1, \quad y_1 = 1.0916.$$

To find $y(x_2) = y_2$:-

$$\therefore y(x_1 + h) = y_2^{(1)} = y_1 + hf(x_1, y_1)$$

$$= 1.0916 + (0.1) f(0.1, 1.0916)$$

$$= 1.0916 + (0.1) (0.8333)$$

$$\Rightarrow y_2^{(1)} = 1.1748$$

$$\therefore y_2^{(2)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})]$$

$$= 1.0916 + \frac{0.1}{2} [f(0.1, 1.0916) + f(0.2, 1.1748)]$$

$$= 1.0916 + 0.05 [(0.8333) + 0.7090]$$

$$\Rightarrow y_2^{(2)} = 1.1687.$$

$$\therefore y_2^{(3)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(2)})]$$

$$= 1.0916 + \frac{0.1}{2} [f(0.1, 1.0916) + f(0.2, 1.1687)]$$

$$= 1.0916 + 0.05 [(0.8322) + 0.7078]$$

$$\Rightarrow y_2^{(3)} = 1.1686$$

$$\therefore y(0.2) = 1.1686.$$

4. Solve $\frac{dy}{dx} = x + \sin y$, $y(0) = 1$. Employ $y(0.4)$ by taking $h = 0.2$ using modified Euler's method.

Sol.

$$\frac{dy}{dx} = x + \sin y = y'$$

$$\& y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$$

To find $y(x_1) = y_1$:-

$$\begin{aligned} \therefore y(x_0 + h) &= y_1 = y_0 + hf(x_0, y_0) \\ &= 1 + 0.2f(0, 1) \\ &= 1 + 0.2(0.8415) \end{aligned}$$

$$\Rightarrow y_1^{(1)} = 1.1683$$

$$\begin{aligned} \therefore y_1^{(2)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})] \\ &= 1 + \frac{0.2}{2} [(0.8415) + f(0.2, 1.1683)] \\ &= 1 + 0.1 [0.8415 + 1.1301] \end{aligned}$$

$$\Rightarrow y_1^{(2)} = 1.1962$$

$$\begin{aligned} \therefore y_1^{(3)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})] \\ &= 1 + \frac{0.2}{2} [(0.8415) + f(0.2, 1.1962)] \\ &= 1 + 0.1 [0.8415 + 1.1307] \end{aligned}$$

$$\Rightarrow y_1^{(3)} = 1.1972.$$

$$\begin{aligned}\therefore y_1^{(4)} &= y_0 + \frac{h}{2} [f(0,1) + f(x_1, y_1^{(3)})] \\ &= 1 + 0.1 [(0.8415) + f(0.2, 1.1973)] \\ &= 1 + 0.1973\end{aligned}$$

$$\Rightarrow y_1^{(4)} = 1.1973$$

$$\therefore y(0.2) = 1.1973, \quad x_1 = 0.2, \quad y_1 = 1.1973$$

To find $y(x_2) = y_2$:-

$$\begin{aligned}\therefore y(x_2) &= y_2^{(1)} = y_1 + hf(x_1, y_1) \\ &= 1.1973 + (0.2)f(0.2, 1.1973) \\ &= 1.1973 + 0.2(1.1311) \\ \Rightarrow y_2^{(1)} &= 1.4235\end{aligned}$$

$$\begin{aligned}\therefore y_2^{(2)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})] \\ &= 1.1973 + \frac{0.2}{2} [f(0.2, 1.1973) + f(0.4, 1.4235)] \\ &= 1.1973 + 0.1 [1.1311 + 1.3892] \\ \Rightarrow y_2^{(2)} &= 1.4493\end{aligned}$$

$$\begin{aligned}\therefore y_2^{(3)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(2)})] \\ &= 1.1973 + \frac{0.2}{2} [f(0.2, 1.1973) + f(0.4, 1.4493)] \\ \Rightarrow y_2^{(3)} &= 1.4497\end{aligned}$$

$$\begin{aligned}\therefore y_2^{(4)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(3)})] \\ &= 1.1973 + 0.1 [f(0.2, 1.1973) + f(0.4, 1.4497)] \\ &= 1.1973 + 0.1 [1.1311 + 1.3927] \\ \Rightarrow y_2^{(4)} &= 1.4497\end{aligned}$$

5. Use modified Euler's methods to compute $y(20.2)$, $y(20.4)$ given $\frac{dy}{dx} = \log_{10}\left(\frac{x}{y}\right)$ with $y(20) = 5$ taking $h = 0.2$.

sol:- Given, $\frac{dy}{dx} = \log_{10}\left(\frac{x}{y}\right)$

& $y(20) = 5 \Rightarrow x_0 = 20, y_0 = 5, h = 0.2$

To find $y(x_1) = y_1$:-

$$\begin{aligned}\therefore y(x_0 + h) &= y_1^{(1)} = y_0 + hf(x_0, y_0) \\ &= 5 + (0.2)f(20, 5) \\ &= 5 + (0.2)(0.6031) \\ \Rightarrow y_1^{(1)} &= 5.1204\end{aligned}$$

$$\begin{aligned}\therefore y_1^{(2)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})] \\ &= 5 + \frac{0.2}{2} [f(20, 5) + f(20.2, 5.1204)] \\ &= 5 + 0.1 [0.6031 + 0.596] \\ \Rightarrow y_1^{(2)} &= 5.1198\end{aligned}$$

$$\begin{aligned}\therefore y_1^{(3)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})] \\ &= 5 + \frac{0.2}{2} [f(20, 5) + f(20.2, 5.1198)] \\ &= 5 + 0.1 [0.6031 + 0.5961] \\ \Rightarrow y_1^{(3)} &= 5.1198\end{aligned}$$

$$\therefore y(20.2) = 5.1198$$

$$\Rightarrow x_1 = 20.2, y_1 = 5.1198$$

To find $y(x_2) = y_2$:-

$$\begin{aligned}\therefore y(x_1+h) &= y_2^{(1)} = y_1 + hf(x_1, y_1) \\ &= 5.1198 + (0.2)f(20.2, 5.1198) \\ &= 5.1198 + (0.2)(0.5961) \\ \Rightarrow y_2^{(1)} &= 5.239.\end{aligned}$$

$$\begin{aligned}\therefore y_2^{(2)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})] \\ &= 5.1198 + \frac{0.2}{2} [f(20.2, 5.1198) + f(20.4, 5.239)] \\ &= 5.1198 + 0.1 [0.5961 + 0.5904]\end{aligned}$$

$$\Rightarrow y_2^{(2)} = 5.2385$$

$$\begin{aligned}\therefore y_2^{(3)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(2)})] \\ &= 5.1198 + 0.1 [f(20.2, 5.1198) + f(20.4, 5.2384)] \\ \Rightarrow y_2^{(3)} &= 5.2385\end{aligned}$$

$$\therefore y(20.4) = 5.2385$$

6. Using modified Euler's method to find $y(0.1)$ given $\frac{dy}{dx} = x^2 + y = f(x, y)$, $y(0) = 1$, perform three iterations by taking $h = 0.05$.

Sol: Given, $\frac{dy}{dx} = x^2 + y = f(x, y)$

$$y(0) = 1, x_0 = 0, y_0 = 1, h = 0.05.$$

To find $y(x_1) = y_1$:-

$$\begin{aligned}\therefore y(x_0+h) &= y_1^{(1)} = y_0 + hf(x_0, y_0) \\ &= 1 + (0.05)f(0, 1) \\ &= 1 + (0.05)(-1)\end{aligned}$$

$$\Rightarrow y_1^{(1)} = 0.95$$

$$\begin{aligned}\therefore y_1^{(2)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})] \\ &= 1 + \frac{0.05}{2} [f(0, 1) + f(0.05, 0.95)]\end{aligned}$$

$$\Rightarrow y_1^{(2)} = 0.9513$$

$$\begin{aligned}\therefore y_1^{(3)} &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})] \\ &= 1 + \frac{0.05}{2} [f(0, 1) + f(0.05, 0.9513)]\end{aligned}$$

$$\Rightarrow y_1^{(3)} = 0.9513$$

$$\therefore y(0.05) = 0.9513$$

$$\Rightarrow x_1 = 0.05, y_1 = 0.9513$$

To find $y(x_2) = y_2$:-

$$\begin{aligned}\therefore y(x_1 + h) &= y_2^{(1)} = y_1 + hf(x_1, y_1) \\ &= 0.9513 + (0.05)f(0.05, 0.9513) \\ &= 0.9513 + (0.05)(-0.9488)\end{aligned}$$

$$\Rightarrow y_2^{(1)} = 0.9039$$

$$\begin{aligned}\therefore y_2^{(2)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})] \\ &= 0.9513 + \frac{0.05}{2} [f(0.05, 0.9513) + f(0.1, 0.9039)]\end{aligned}$$

$$\Rightarrow y_2^{(2)} = 0.9052$$

$$\begin{aligned}\therefore y_2^{(3)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(2)})] \\ &= 0.9513 + \frac{0.05}{2} [f(0.05, 0.9513) + f(0.1, 0.9052)]\end{aligned}$$

$$\Rightarrow y_2^{(3)} = 0.9052$$

$$\therefore y(0.1) = 0.9052$$

7. Using Euler's modified method, obtain a solution of the equation $\frac{dy}{dx} = x + |\sqrt{y}|$ with initial condition $y=1$ & $x=0$ for the range $0 \leq x \leq 0.4$ in step of 0.2.

Sol: Given, $\frac{dy}{dx} = x + |\sqrt{y}| = f(x, y)$, $h=0.2$

and $y(0)=1 \Rightarrow x_0=0, y_0=1$

To find $y(x_1)=y_1$:-

$$\therefore y(x_0+h) = y_1^{(1)} = y_0 + hf(x_0, y_0) \\ = 1 + (0.2)f(0, 1)$$

$$\Rightarrow y_1^{(1)} = 1.2$$

$$\therefore y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})] \\ = 1 + \frac{0.2}{2} [f(0, 1) + f(0.2, 1.2)]$$

$$\Rightarrow y_1^{(2)} = 1.2295$$

$$\therefore y_1^{(3)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})] \\ = 1 + \frac{0.2}{2} [f(0, 1) + f(0.2, 1.2295)]$$

$$\Rightarrow y_1^{(3)} = 1.2309$$

$$\therefore y_1^{(4)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(3)})] \\ = 1 + \frac{0.2}{2} [f(0, 1) + f(0.2, 1.2309)]$$

$$\Rightarrow y_1^{(4)} = 1.2309$$

$$\therefore y_1^{(4)} = 1.2309$$

$$\therefore y(0.2) = 1.2309$$

$$\Rightarrow x_1 = 0.2, y_1 = 1.2309$$

To find $y(x_2) = y_2$:-

$$\begin{aligned}\therefore y(x_1 + h) &= y_2^{(1)} = y_1 + hf(x_1, y_1) \\ &= 1.2309 + 0.2f(0.2, 1.2309)\end{aligned}$$

$$\Rightarrow y_2^{(1)} = 1.4928$$

$$\begin{aligned}\therefore y_2^{(2)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_1, y_2^{(1)})] \\ &= 1.2309 + \frac{0.2}{2} [f(0.2, 1.2309) + f(0.4, 1.4928)] \\ &= 1.2309 + 0.1 [1.3095 + 1.6218]\end{aligned}$$

$$\Rightarrow y_2^{(2)} = 1.524$$

$$\begin{aligned}\therefore y_2^{(3)} &= y_1 + \frac{h}{3} [f(x_1, y_1) + f(x_2, y_2^{(2)})] \\ &= 1.2309 + \frac{0.2}{3} [f(0.2, 1.2309) + f(0.4, 1.524)]\end{aligned}$$

$$\Rightarrow y_2^{(3)} = 1.5253$$

$$\begin{aligned}\therefore y_2^{(4)} &= y_1 + \frac{h}{4} [f(x_1, y_1) + f(x_2, y_2^{(3)})] \\ &= 1.2309 + \frac{0.2}{4} [f(0.2, 1.2309) + f(0.4, 1.5253)] \\ &= 1.2309 + 0.1 [1.3095 + 1.635]\end{aligned}$$

$$\Rightarrow y_2^{(4)} = 1.5254$$

$$\therefore y_2(0.4) = 1.5254.$$

Runge - Kutta method of fourth order:-

Step-1:- Write given D.E into standard form

$$\frac{dy}{dx} = f(x, y) \text{ to the initial condition } y(x_0) = y_0.$$

Step-2:- To find $y(x_1) = y_1$, the R-K method follows

$$y(x_0 + h) = y(x_1) = y_0 + \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4]$$

$$\text{Where, } K_1 = hf(x_0, y_0)$$

$$K_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right)$$

$$K_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right)$$

$$K_4 = hf(x_0 + h, y_0 + K_3)$$

1. Using Runge-kutta method of fourth order, find y at $x = 0.1$, given $\frac{dy}{dx} = 3e^x + 2y$, $y(0) = 1$.

Sol:- Given, $\frac{dy}{dx} = 3e^x + 2y = f(x, y)$

and $y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$, let $h = 0.1$

$$\therefore K_1 = hf(x_0, y_0) = (0.1)f(0, 1)$$
$$= (0.1)(5)$$

$$\Rightarrow K_1 = 0.5$$

$$\therefore K_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right)$$
$$= (0.1)f(0.05, 1.25)$$
$$= (0.1)(5.6538)$$

$$\Rightarrow K_2 = 0.5638$$

$$\therefore K_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right)$$

$$= (0.1) f(0.05, 1.2827)$$

$$= (0.1) (5.7192)$$

$$\Rightarrow K_3 = 0.57192$$

$$\therefore K_4 = hf(x_0 + h, y_0 + K_3)$$

$$= (0.1) f(0.1, 1.57192)$$

$$= (0.1) (6.4593)$$

$$\Rightarrow K_4 = 0.64593$$

WKT,

$$y(x_1) = y_0 + \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4]$$

$$\Rightarrow y(0.1) = 1 + \frac{1}{6} [0.5 + 1.13076 + 1.14384 + 0.64593]$$

$$= 1 + \frac{1}{6} (3.42053)$$

$$\Rightarrow y(0.1) = 1.5700$$

2. Using Runge-kutta method of fourth order to find $y(0.1)$, if $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2} = f(x, y)$ with $y(0) = 1$.

sol: Given, $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2} = f(x, y)$

$$y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1 \text{ \& } h = 0.1$$

$$\therefore K_1 = hf(x_0, y_0)$$

$$= (0.1) f(0, 1)$$

$$= (0.1) (1)$$

$$\Rightarrow K_1 = 0.1$$

$$\begin{aligned}\therefore K_2 &= hf \left(x_0 + \frac{0.1}{2}, y_0 + \frac{K_1}{2} \right) \\ &= (0.1) f(0.05, 1.05) \\ &= (0.1) (0.9954)\end{aligned}$$

$$\Rightarrow K_2 = 0.09954$$

$$\begin{aligned}\therefore K_3 &= (0.1) f(0.05, 1.0498) \\ &= (0.1) (0.9955)\end{aligned}$$

$$\Rightarrow K_3 = 0.09955$$

$$\begin{aligned}\therefore K_4 &= (0.1) f(0.1, 1.09955) \\ &= (0.1) (0.9836)\end{aligned}$$

$$\Rightarrow K_4 = 0.09836$$

$\therefore$ WKT,

$$y(x_0 + h) = y(x_1) = y_0 + \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4]$$

$$\Rightarrow y(0.1) = 1 + \frac{1}{6} [0.1 + 0.1991 + 0.1991 + 0.09836]$$

$$\Rightarrow y(0.1) = 1 + \frac{1}{6} [0.5966]$$

$$\Rightarrow y(0.1) = 1.0994$$

3. Using Runge-Kutta method, find $y(0.2)$, give that

$$\frac{dy}{dx} = \frac{y-x}{y+x}, \quad y(0) = 1.$$

Sol: Given $\frac{dy}{dx} = \frac{y-x}{y+x} = f(x, y)$

$$\text{and } y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1, h = 0.2$$

$$\begin{aligned}\therefore K_1 &= hf(x_0, y_0) = 0.2 f(0, 1) \\ &= 0.2 (1)\end{aligned}$$

$$\Rightarrow K_1 = 0.2$$

$$\therefore K_2 = hf \left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2} \right)$$

$$= 0.2 f(0.1, 1.1)$$

$$= 0.2(0.8333)$$

$$\Rightarrow K_2 = 0.1667$$

$$\therefore K_3 = hf \left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2} \right)$$

$$= 0.2 f(0.1, 1.0834)$$

$$= 0.2(0.8310)$$

$$\Rightarrow K_3 = 0.1662$$

$$\therefore K_4 = hf(x_0 + h, y_0 + K_3)$$

$$= 0.2 f(0.2, 1.1662)$$

$$= 0.2(0.7072)$$

$$\Rightarrow K_4 = 0.1414$$

WKT,

$$y(x_1) = y_0 + \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4]$$

$$y(0.2) = 1 + \frac{1}{6} [0.2 + 2(0.1667) + 2(0.1662) + 0.1414]$$

$$= 1 + \frac{1}{6} [0.2 + 0.3334 + 0.3324 + 0.1414]$$

$$= 1 + \frac{1}{6} [1.0072]$$

$$= 1 + 0.1679$$

$$\Rightarrow y(0.2) = 1.1679$$

4. Using Runge-Kutta method, find $y(1.1)$ given that $\frac{dy}{dx} + y = 2x$ and $y(1) = 3$.

Sol: Given, $\frac{dy}{dx} = 2x - y = f(x, y)$,

$$\& \ y(1) = 3 \Rightarrow x_0 = 1, y_0 = 3$$

$$\therefore k_1 = hf(x_0, y_0) = 0.1 f(1, 3) \\ = 0.1(-1)$$

$$\Rightarrow k_1 = -0.1$$

$$\therefore k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \\ = 0.1 f(1.05, 2.95) \\ = 0.1(-0.85)$$

$$\Rightarrow k_2 = -0.0850$$

$$\therefore k_3 = hf(x_0 + h, y_0 + k_2) \\ = 0.1 f(1.1, 2.9149) \\ = 0.1(-0.7149)$$

$$\Rightarrow k_3 = -0.0714$$

WKT,

$$y(x_1) = 3 + \frac{1}{6} [-0.1 + 2(-0.0850) + 2(-0.0858) - 0.0714]$$

$$y(1.1) = 3 + \frac{1}{6} [-0.1 - 0.1700 - 0.1716 - 0.0714]$$

$$= 3 + \frac{1}{6} [-0.5130]$$

$$= 3 - 0.0855$$

$$\Rightarrow y(1.1) = 2.9145$$

5. Using Runge-Kutta method, find $y(0.2)$ given that $\frac{dy}{dx} = 3x + \frac{y}{2}$, $y(0) = 1$, $h = 0.2$.

sol:- Given, $\frac{dy}{dx} = 3x + \frac{y}{2} = f(x, y)$

$$y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1, h = 0.2$$

$$\therefore K_1 = hf(x_0, y_0) = (0.2)f(0, 1) \\ = (0.2)(0.5)$$

$$\Rightarrow K_1 = 0.10$$

$$\therefore K_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right)$$

$$= 0.2f(0.1, 1.05)$$

$$= 0.2(0.8250)$$

$$\Rightarrow K_2 = 0.1650$$

$$\therefore K_3 = hf\left(x_0 + \frac{h}{2}, y_0 + K_2\right)$$

$$= 0.2f(0.1, 1.0825)$$

$$= 0.2(0.8413)$$

$$\Rightarrow K_3 = 0.1683$$

$$\therefore K_4 = hf(x_0 + h, y_0 + K_3)$$

$$= 0.2f(0.2, 1.1683)$$

$$= (0.2)(1.1842)$$

$$\Rightarrow K_4 = 0.2368$$

WKT,

$$y(x_1) = y_0 + \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4]$$

$$y(0.2) = 1 + \frac{1}{6} [0.1 + 0.3300 + 0.3366 + 0.2368]$$

$$= 1 + \frac{1}{6} [0.9434]$$

$$\Rightarrow y(0.2) = 1.1672$$

Milne's Predictor - Corrector method:-

Step-1:- Write the given D.E into the standard form $\frac{dy}{dx} = f(x, y)$, to the initial conditions

$$y(x_0) = y_0, y(x_1) = y_1, y(x_2) = y_2, y(x_3) = y_3.$$

Step-2:- Find $f_0 = f(x_0, y_0)$, $f_1 = f(x_1, y_1)$, $f_2 = f(x_2, y_2)$, $f_3 = f(x_3, y_3)$.

Step-3:- To find $y(x_4) = y_4$, use the following predictor and corrector formulas.

$$y(x_4) = y_4^{(p)} = y_0 + \frac{4h}{3} [2f_1 - f_2 + f_3]$$

and

$$y(x_4) = y_4^{(c)} = y_2 + \frac{h}{3} [f_2 + 4f_3 + f_4^{(p)}].$$

$$\text{where, } y_4^{(p)} = f(x_4, y_4^{(p)})$$

1. Apply Predictor - Corrector method find $y(0.8)$ from $\frac{dy}{dx} = x^3 + y$, given $y(0) = 0$, $y_1(0.2) = 0.073$, $y_2(0.4) = 0.459$, $y_3(0.6) = 3.093$.

Sol: Given, $\frac{dy}{dx} = x^3 + y = f(x, y)$

$$\text{and } y(0) = 0 \Rightarrow x_0 = 0, y_0 = 0$$

$$y_1(0.2) = 0.073 \Rightarrow x_1 = 0.2, y_1 = 0.073$$

$$y_2(0.4) = 0.459 \Rightarrow x_2 = 0.4, y_2 = 0.459$$

$$y_3(0.6) = 3.093 \Rightarrow x_3 = 0.6, y_3 = 3.093$$

$$\therefore f_0 = f(x_0, y_0) = f(0, 0) = 0$$

$$f_1 = f(x_1, y_1) = f(0.2, 0.073) = 0.081$$

$$f_2 = f(x_2, y_2) = f(0.4, 0.459) = 0.516$$

$$f_3 = f(x_3, y_3) = f(0.6, 3.023) = 3.239$$

$$\begin{aligned} \therefore y(x_4) &= y_4^{(p)} = y_0 + \frac{4h}{3} [2f_1 - f_2 + 2f_3] \\ &= 2 + 4 \frac{(0.2)}{3} [4.162 - 2.516 + 6.478] \\ &= 2 + \frac{0.8}{3} (8.124) \end{aligned}$$

$$\Rightarrow y^{(p)}(x_4) = 4.1664 = y(0.8)$$

$$\therefore f_4^{(p)} = f(x_4, y_4^{(p)}) = f(0.8, 4.1664) = 4.6784$$

$$\begin{aligned} \therefore y(x_4) &= y_4^{(c)} = y_2 + \frac{h}{3} [f_2 + 4f_3 + f_4^{(p)}] \\ &= 2.452 + \frac{0.2}{3} [2.516 + 12.956 + 4.6784] \\ &= 2.452 + 1.3434 \end{aligned}$$

$$y_4^{(c)} = 3.7954$$

$$\therefore y(0.8) = 3.7954$$

Q. To find $y(1.4)$ from $\frac{dy}{dx} = x^2 + \frac{y}{x}$, given $y(1) = 2$
 $y_1(1.1) = 2.2156$, $y_2(1.2) = 2.4549$, $y_3(1.3) = 2.7514$.

Sol: Given, $\frac{dy}{dx} = x^2 + \frac{y}{x} = f(x, y)$

$$\& y(1) = 2 \Rightarrow x_0 = 1, y_0 = 2$$

$$y_1(1.1) = 2.2156 \Rightarrow x_1 = 1.1, y_1 = 2.2156$$

$$y_2(1.2) = 2.4549 \Rightarrow y_2 = 1.2, y_2 = 2.4549$$

$$y_3(1.3) = 2.7514 \Rightarrow y_3 = 1.3, y_3 = 2.7514$$

$$\therefore f_0 = f(x_0, y_0) = f(1, 2) = 2$$

$$f_1 = f(x_1, y_1) = f(1.1, 2.3178) = 2.3178$$

$$f_2 = f(x_2, y_2) = f(1.2, 2.6675) = 2.6675$$

$$f_3 = f(x_3, y_3) = f(1.3, 3.0657) = 3.0657$$

$$\therefore y(x_4) = y_4^{(p)} = y_0 + \frac{4h}{3} [2f_1 - f_2 + 2f_3]$$

$$\Rightarrow y_4^{(p)} = 2 + \frac{4(0.1)}{3} [4.6356 - 2.6675 + 6.1314]$$

$$\Rightarrow y_4^{(p)} = 2 + 1.0799$$

$$\Rightarrow y_4^{(p)} = 3.0799 = y(1.4)$$

$$f_4^{(p)} = f(x_4, y_4^{(p)}) = f(1.4, 3.0799) = 3.5$$

$$\therefore y(x_4) = y_4^{(c)} = y_2 + \frac{h}{3} [f_2 + 4f_3 + f_4^{(p)}]$$

$$\Rightarrow y_4^{(c)} = 2.4549 + \frac{0.1}{3} [2.6675 + 12.2628 + 3.5]$$

$$\Rightarrow y_4^{(c)} = 2.4549 + 0.6143$$

$$\Rightarrow y_4^{(c)} = 3.0692$$

3. Given $\frac{dy}{dx} = \frac{(1+x^2)y^2}{2}$, $y(0)=1$, $y_1(0.1)=1.06$,

$y(0.2)=1.12$, $y(0.3)=1.21$, Evaluate $y(0.4)$ using Mile's Predictor - Corrector method.

Sol: Given, $\frac{dy}{dx} = \frac{(1+x^2)y^2}{2}$

and $y(0)=1 \Rightarrow x_0=0, y_0=1$

$$y_1(0.1) = 1.06 \Rightarrow x_1 = 0.1, y_1 = 1.06$$

$$y_2(0.2) = 1.12 \Rightarrow x_2 = 0.2, y_2 = 1.12$$

$$y_3(0.3) = 1.21 \Rightarrow x_3 = 0.3, y_3 = 1.21$$

$$\therefore f_0 = f(0, 1) = 0.5$$

$$f_1 = f(0.1, 1.06) = 0.5644$$

$$f_2 = f(0.2, 1.12) = 0.6523$$

$$f_3 = f(0.3, 1.21) = 0.7979$$

$$\therefore y_4^{(p)} = y_0 + \frac{4h}{3} [2f_1 - f_2 + 2f_3]$$

$$= 1 + \frac{4(0.1)}{3} [1.1348 - 0.6523 + 1.5958]$$

$$= 1 + 0.2771$$

$$y_4^{(p)} = 1.2771$$

$$\therefore f_4^{(p)} = f(x_4, y_4^{(p)}) = f(0.4, 1.2771) = 0.9460$$

$$\therefore y_4^{(c)} = y_2 + \frac{h}{3} [f_2 + 4f_3 + f_4^{(p)}]$$

$$= 1.12 + \frac{0.1}{3} [0.6523 + 3.1916 + 0.9460]$$

$$= 1.12 + 0.1597$$

$$\Rightarrow y_4^{(c)} = 1.2797$$

$$y_4(0.4) = 1.2797$$

4. Use Milne's predictor-corrector method to find $y(4.5)$, given $\frac{dy}{dx} = \frac{2-y^2}{5x}$, given $y(4.1) = 1.0049$, $y(4.2) = 1.0097$, $y(4.3) = 1.0143$, $y(4.4) = 1.0187$.

Sol: Given, $\frac{dy}{dx} = \frac{2-y^2}{5x} = f(x, y)$

$$\text{and } y(4.1) = 1.0049 \Rightarrow x_0 = 4.1, y_0 = 1.0049$$

$$y(4.2) = 1.0097 \Rightarrow x_1 = 4.2, y_1 = 1.0097$$

$$y(4.3) = 1.0143 \Rightarrow x_2 = 4.3, y_2 = 1.0143$$

$$y(4.4) = 1.0187 \Rightarrow x_3 = 4.4, y_3 = 1.0187$$

$$\therefore f(4.1, 1.0049) \Rightarrow f_0 = 0.0483$$

$$f_1(4.2, 1.0097) \Rightarrow f_1 = 0.0467$$

$$f_2(4.3, 1.0143) \Rightarrow f_2 = 0.0452$$

$$f_3(4.4, 1.0187) \Rightarrow f_3 = 0.0437$$

$$\therefore y_4^{(p)} = y_0 + \frac{4h}{3} [2f_1 - f_2 + 2f_3]$$

$$= 1.0049 + \frac{4(0.1)}{3} [0.0934 - 0.0452 + 0.0874]$$

$$= 1.0049 + 0.0181$$

$$\Rightarrow y_4^{(p)} = 1.023$$

$$\therefore f_4^{(p)} = f(4.5, 1.023) = 0.0424$$

$$\therefore y_4^{(c)} = y_2 + \frac{h}{3} [f_2 + 4f_3 + f_4^{(p)}]$$

$$= 1.0143 + \frac{(0.1)}{3} [0.0452 + 0.1743 + 0.0424]$$

$$= 1.0143 + 0.0087$$

$$\Rightarrow y_4^{(0)} = 1.023$$

$$\therefore y(0.4) = 1.023$$

5. Use Milne's method to compute $y(0.4)$ from $\frac{dy}{dx} = x - y^2$ given $y(0) = 1$, $y(0.1) = 0.9117$, $y(0.2) = 0.8494$, $y(0.3) = 0.8061$.

Sol: Given, $\frac{dy}{dx} = x - y^2 = f(x, y)$

$$\text{and } y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$$

$$y(0.1) = 0.9117 \Rightarrow x_1 = 0.1, y_1 = 0.9117$$

$$y(0.2) = 0.8494 \Rightarrow x_2 = 0.2, y_2 = 0.8494$$

$$y(0.3) = 0.8061 \Rightarrow x_3 = 0.3, y_3 = 0.8061$$

$$\therefore y_4^{(p)} = y_0 + \frac{4h}{3} [2f_1 - f_2 + 2f_3]$$

$$= 1 + \frac{4(0.1)}{3} [-1.4624 + 0.5215 - 0.6996]$$

$$\Rightarrow y_4^{(p)} = 1 - 0.2187 = 0.7813$$

$$\therefore f_4^{(p)} = f(0.4, 0.7813) = -0.2104$$

$$\therefore y_4^{(c)} = y_2 + \frac{h}{3} [f_2 + 4f_3 + f_4^{(p)}]$$

$$= 0.8494 + \frac{(0.1)}{3} [-0.5215 - 1.3992 - 0.2104]$$

$$= 0.8494 - 0.0710$$

$$\Rightarrow y_4^{(c)} = 0.7784$$

$$\therefore y(0.4) = 0.7784$$